SECURING

AGENTIC AI SYSTEMS

A Short Workshop

ISACA Omaha Chapter Spring Seminar

Christopher Williams

From ML Fundamentals to Agentic Defense

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Omaha Chapter

YOUR SPEAKER

CHRISTOPHER WILLIAMS

- Principal Incident Response Engineer
- Active CISSP · CISM · GCFA
- B.S. in Cyber Security from Bellevue University
- Graduate Student at Bellevue
- 30+ years industry experience
- Linux/Unix, DevOps, Ethical Hacker

My presentation, comments and opinions are provided in my personal capacity and not as a representative of my employer. They do not reflect the views of my employer and are not endorsed by my employer.

WORKSHOP REQUIREMENTS

What you need to run the labs locally

HARDWARE OPTIONS

- Linux with modern Nvidia GPU

— OR —

- Windows / Mac with hypervisor:

VMware Workstation / Fusion

VirtualBox

- Ollama-compatible system accepted

TOOLS USED

Python · PyTorch · Transformers · Ollama · Open WebUI · FastMCP · LangChain · LangGraph · OpenClaw

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AGENDA

1 hour · 5 sections · Agentic AI Defense

Lab 1	Building Agents	10 min
Lab 2	Autonomous Agents	10 min
Lab 3	Attacking LLMs	10 min
Lab 4	Detection & Alerting	10 min
Lab 5	Canaries In The Coalmine	10 min

Q&A · Open Discussion

LAB 1: BUILDING AGENTS

LECTURE

- What is an AI agent?
- Agents vs chatbots vs pipelines
- The ReAct pattern: Reason + Act
- LangChain: chains, agents, runnables
- Agent components: LLM, tools, memory
- Execution loops and state management
- When do agents go wrong?

LAB EXERCISE

- Build an agent using LangChain

LAB 2: AUTONOMOUS AGENTS

LECTURE

- Intro to OpenClaw framework
- LangGraph: stateful agentic graphs
- Multi-agent coordination patterns
- Goal decomposition and planning
- Persistence, checkpoints, memory
- Agentic attack surface expansion
- Real-world autonomous agent risks

LAB EXERCISE

- Launch OpenClaw with an objective
- Give the agent a multi-step mission
- Observe autonomous planning behavior

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LAB 3: ATTACKING LLMs

Prompt Injection

Force unintended behavior via crafted input

Jailbreaking

Bypass safety guardrails and policies

Data Exfiltration

Extract training data or context window

Poisoning Attacks

Corrupt training or RAG data sources

Social Engineering

Manipulate users through AI output

Insecure Output

Exploit downstream processing of LLM output

Multimodal Attacks

Image or audio as attack vector

Tool / Agent Abuse

Invoke tools for unintended actions

Supply Chain

Compromise models, plugins, or MCP servers

DoS & Hybrid

Resource exhaustion and combined techniques

LAB 4: DETECTION AND ALERTING

LECTURE

- Needle in a haystack: what matters?
- Signal vs noise in AI system logs
- Generating alerts from log events
- Rule writing and alert correlation

LAB EXERCISE

- Discuss Alerting Rules
- Discuss: alert fatigue vs blind spots

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LAB 5: CANARIES IN THE COAL MINE

LECTURE

- What are canary tokens?
- Lure attackers · detect access
- Token types: URLs, files, credentials
- Planting tokens in AI contexts:
  - System prompts
  - Tool responses
  - RAG document stores
- LLM data exfil detection via canaries

LAB EXERCISE

- Plant canary tokens in the environment
- Embed in documents & tool outputs
- Discuss use cases:
  - RAG poisoning detection
  - LLM-based exfil detection
  - Insider threat identification

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QUESTIONS?

Thank you for attending!

Christopher Williams
Email: gnulinuxadmin@gmail.com
Signal: cybersquirrel.73
Discord: cybersquirrelfl

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All materials are open source. Labs, slides, and code will be shared. Based on an 8hr workshop at BSidesOK. You can download and run all the labs on your own time and experiment with them.

https://github.com/gnulinuxadmin/HackNWA-AI-For-Ethical-Hackers

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